

Ch. 11 — P03 (A) 2² Factorial Effects

Problem Bank | Chapter 11 | Type: A | Difficulty:

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A simulation experiment studies a call centre with two factors:

- **Factor A:** Number of agents (1 or 3)
- **Factor B:** Maximum queue length before balking (10 or 50)

Response: mean call wait time W_q (minutes). Each cell is the average of $r = 10$ replications.

	B = 10 (low)	B = 50 (high)
A = 1 (low)	8.4	12.1
A = 3 (high)	1.6	2.8

Tasks

1. Compute the main effect of A (number of agents). Interpret: what does this tell you about the impact of adding agents?
2. Compute the main effect of B (queue length limit). Interpret: why does a longer queue (more balking room) increase W_q ?
3. Compute the AB interaction. Draw an interaction plot. Are the effects additive? What does the interaction tell you about the system?
4. Fit the linear model:

$$W_q = \mu + \alpha x_A + \beta x_B + \gamma x_A x_B$$

where $x_A, x_B \in \{-1, +1\}$. Solve for $\hat{\mu}, \hat{\alpha}, \hat{\beta}, \hat{\gamma}$.

5. Use your model to predict W_q at the centre point $(x_A, x_B) = (0, 0)$ (i.e., 2 agents, queue limit = 30). What is the limitation of using a linear model to make this prediction?
6. If the call centre manager can implement only one change (either adding 2 agents *or* increasing the queue limit from 10 to 50), which should they choose? Base your answer on the factorial effects.